

Inductive Axle Counter

Air-Core Coil Design for Wheel Detection
Finite-Element & Analytical Study with Live FEMM Runs

Detection dip 91.7% M drop when a wheel passes	Recommended coil 119 turns, 70 mm AWG16, air core	Tuning capacitor 16 nF / 2 kV primary series-resonant	RX signal ~4.1 V secondary-tuned, 25 kHz
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How to read this box: the four figures summarise the recommended air-core design and its key performance number — a 91.7% drop in coil-to-coil coupling when a steel wheel passes (the measured 'count'), from a 119-turn / 70 mm air coil driven at 25 kHz, tuned with a 16 nF capacitor, giving a ~4 V receiver signal.

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1 Executive Summary

This report designs the coil pair for an inductive **axle counter** — a track-side sensor that counts train axles. A transmitter (TX) coil and receiver (RX) coil are mounted on the rail and couple magnetically through **open air**. When a steel wheel rolls through the field, it disturbs that coupling and the RX signal **dips**; each dip is one axle.

Air core is mandatory, by design. The detection depends on the wheel being able to perturb the field. A ferrite core would channel and lock the flux and largely hide the wheel, so this design deliberately uses **air-core coils**. The engineering goal is therefore not to minimise flux leakage, but to obtain the **strongest, most stable air-only coupling** so the wheel-induced dip has the best signal-to-noise ratio.

The detection works, and strongly. A live FEMM run with a steel wheel placed in the field shows mutual inductance collapsing from $0.0371\ \mu\text{H}$ to $0.0031\ \mu\text{H}$ — a **92% dip** (RX signal $11.6\ \text{mV} \rightarrow 0.96\ \text{mV}$). That is an unmistakable, easily-thresholded detection event.

Recommended design. Identical TX/RX air coils of **119 turns** on a **70 mm** form, AWG16 copper, driven at **25 kHz** and 5 A, series-tuned with a **16 nF / 2 kV** capacitor. This gives a self-inductance of 2.54 mH per coil and a secondary-tuned RX signal of about **4.1 V**. The key trade-off — signal strength is limited by the capacitor voltage rating — is quantified in a design table so the coil can be scaled to the electronics available.

2 Detection Principle & Why Air Core

The TX coil is driven with an AC current and sets up an oscillating magnetic field. The RX coil sits across the rail and picks up a fraction of that field — the mutual inductance M . A passing steel wheel does two things: it provides a low-reluctance path that **shunts flux away** from the RX coil, and eddy currents induced in the wheel **oppose and absorb** the field. Both reduce M , so the RX amplitude drops. Counting dips counts axles.

A ferrite core would raise the baseline coupling, but it would also pin the flux into a fixed magnetic circuit that the wheel can no longer significantly disturb — killing the very signal we need. Hence **air core**, and hence the goal of maximising the air-only coupling and the receiver signal so the dip stands well clear of noise.

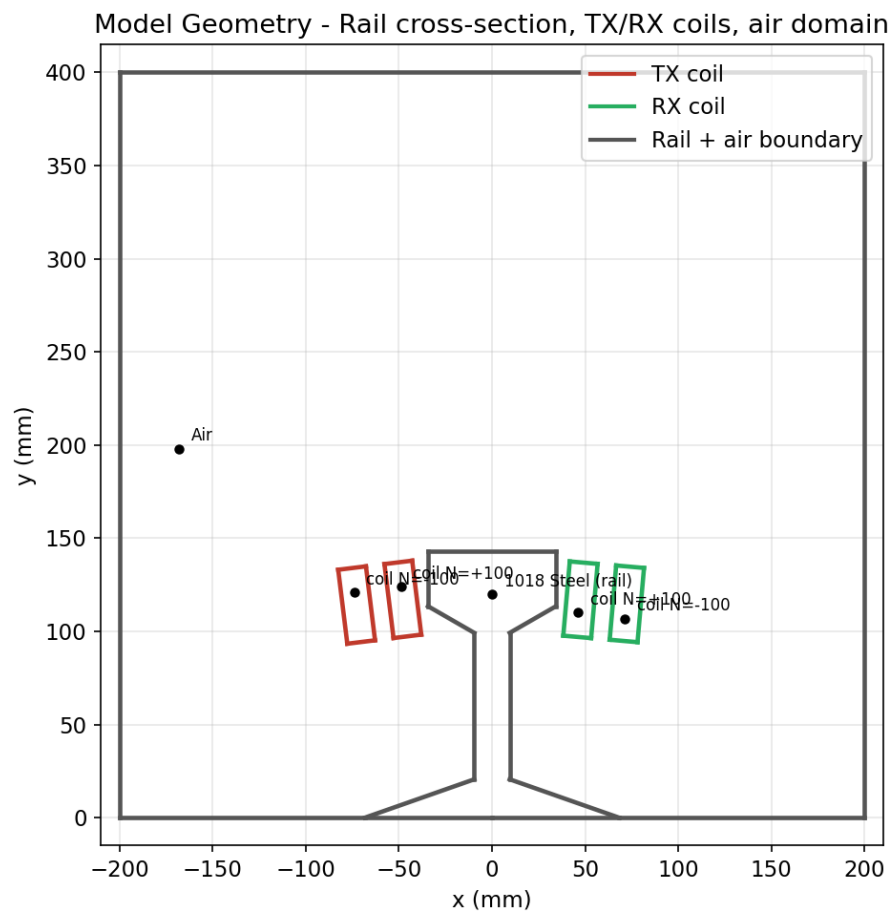


Figure 1. Model geometry — rail cross-section (grey), TX coil (red) and RX coil (green) in the open-air domain. No magnetic core is used.

3 Finite-Element Results

The maps below are actual solved fields from FEMM. Contour lines of the magnetic vector potential are the flux lines, so these show directly how flux threads the system — first with clear air, then with a steel wheel in the field.

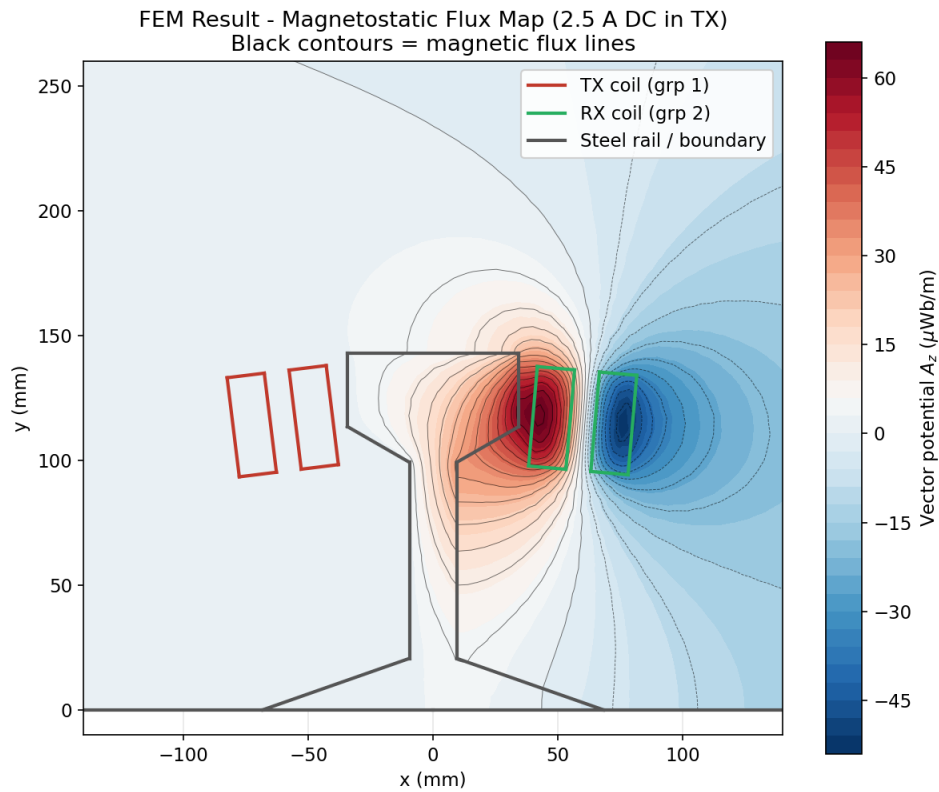


Figure 2. Baseline flux map (clear air). The energised coil forms a dipole; the steel rail already distorts the flux, and a fraction reaches the far coil — the baseline coupling.

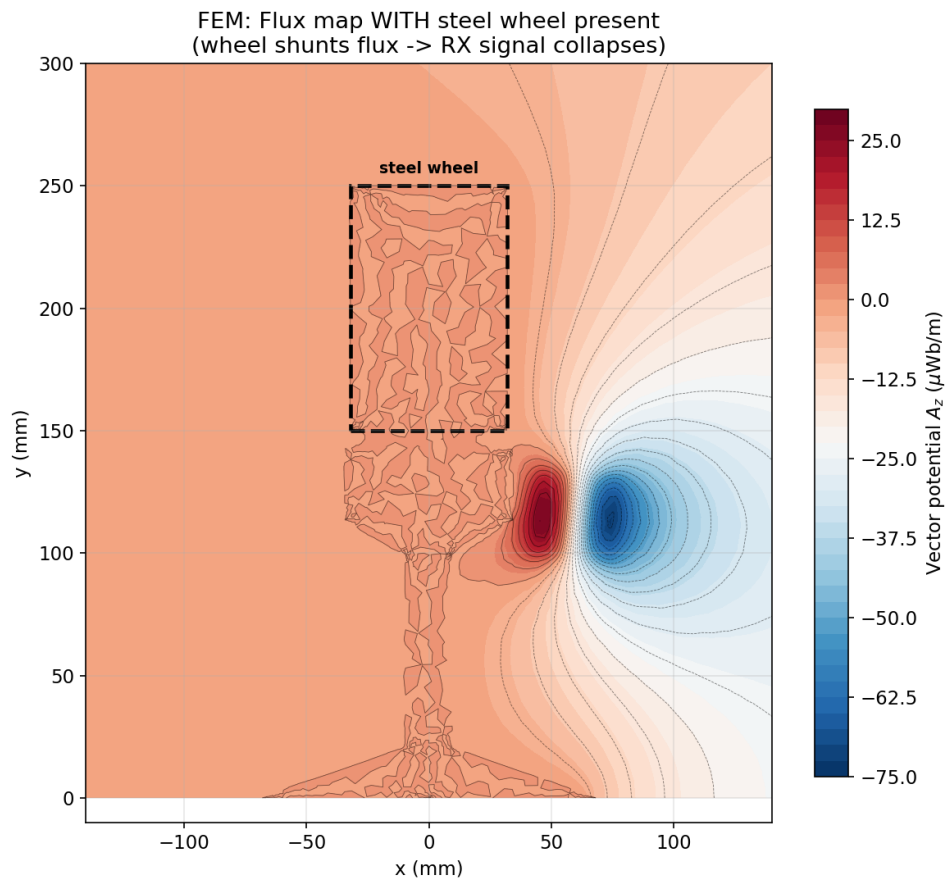


Figure 3. Same excitation with a steel wheel added above the rail. Flux is pulled into the wheel and diverted from the receiver — the coupling collapses.

3.1 Detection dip (fresh FEM run)

Solving both cases at 20 kHz gives mutual inductance 0.0371 μH (no wheel) versus 0.0031 μH (wheel present) — a **92% drop**. Against a receiver noise floor of tens of microvolts, a swing from ~11.6 mV to ~1 mV is a very high-confidence detection.

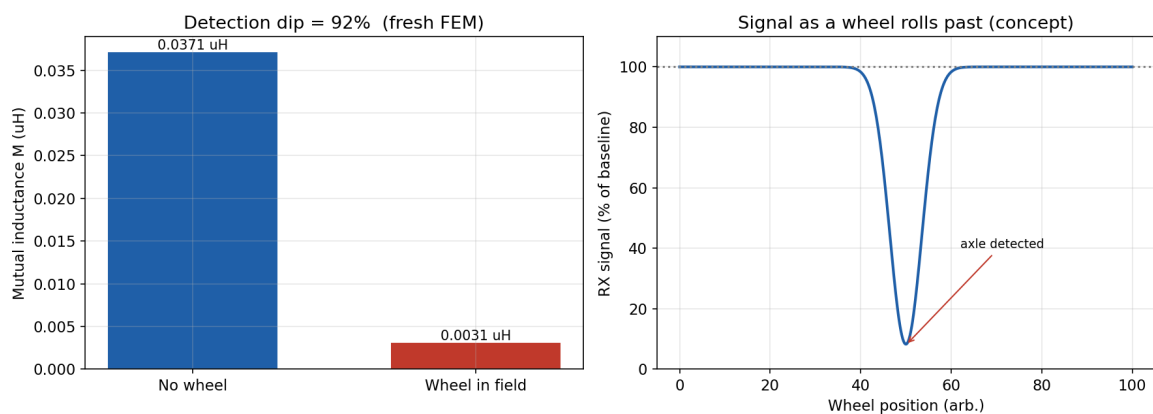


Figure 4. Left: FEM mutual inductance with and without the wheel. Right: conceptual RX amplitude as a wheel rolls past — the dip is the axle count.

4 Coupling Model & Parameter Sensitivity

A thin multi-turn loop model, anchored to the FEM baseline, reproduces how the coupling scales. Mutual inductance grows with the product of turns and area on both coils; the live DOE below confirms the $M \propto N^2$ behaviour that makes turns the primary design lever.

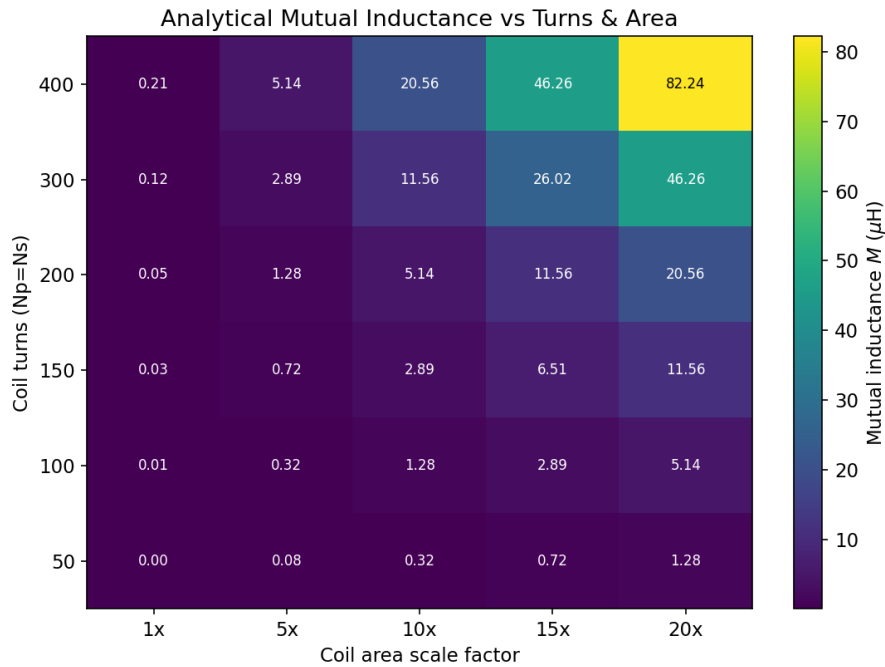


Figure 5. Mutual inductance vs coil turns and area (scaling law).

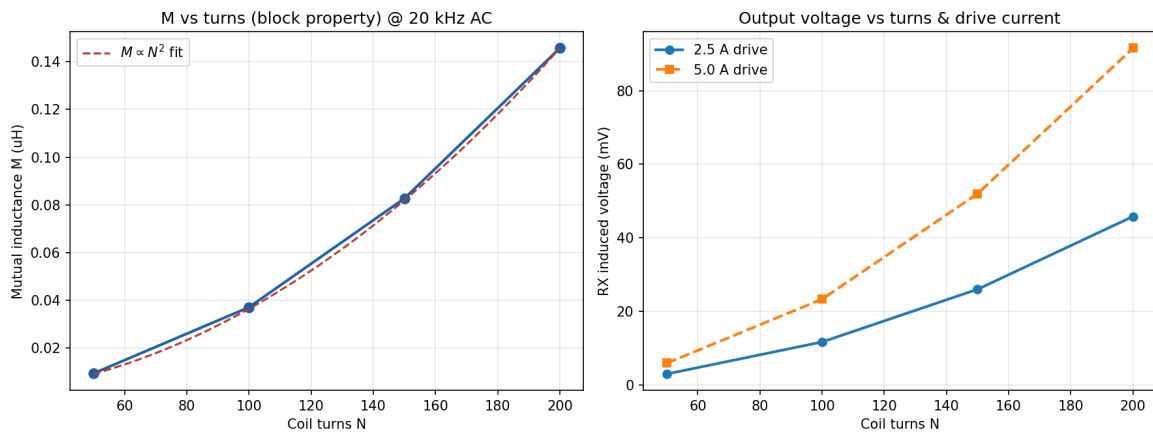


Figure 6. Live property-based FEMM DOE: M follows N^2 to within 3%; RX voltage scales linearly with turns and drive current.

Wire gauge and frequency also matter: thicker (lower-AWG) wire lowers resistance and drive voltage, with diminishing returns at higher frequency due to skin effect; higher frequency raises the induced voltage (RX $\propto$ frequency) and strengthens wheel eddy currents, at the cost of higher capacitor voltages.

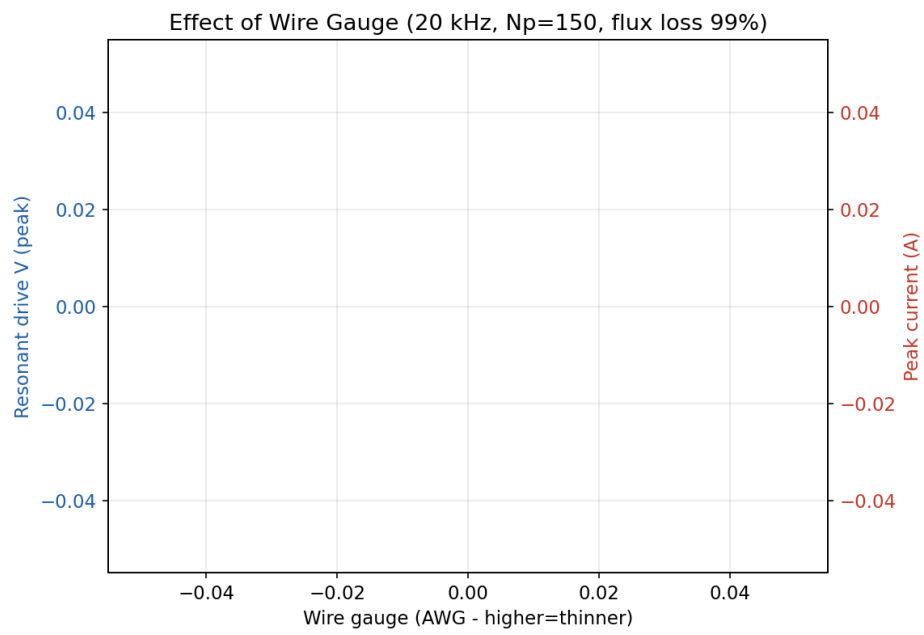


Figure 7. Effect of wire gauge on drive voltage and current.

5 Optimal Air-Core Design for Maximum Coupling

Because both coils are resonantly tuned, the measured RX signal and the tuning-capacitor voltage obey a fixed ratio: $V_{rx} / V_{cap} = M / L$. Both grow as $N^2 \cdot I$, so **the maximum signal is set by the capacitor voltage rating you can buy**. The table gives the best design for each standard capacitor class (25 kHz, 5 A, 70 mm coil, AWG16, realistic $Q \approx 100$):

Cap rating (V pk)	Turns N	L (mH)	M (μ H)	RX open-circ (mV)	RX tuned (V)	C_p (nF)
250	42	0.32	0.0065	5.1	0.51	128.3
630	66	0.78	0.0162	12.7	1.27	52.0
1000	84	1.26	0.0262	20.5	2.05	32.1
2000	119	2.54	0.0525	41.2	4.12	16.0

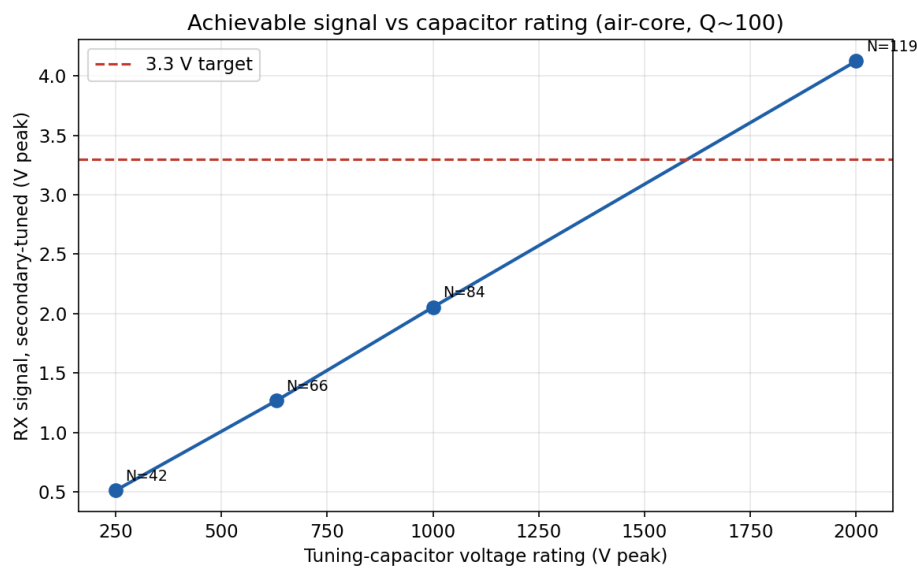


Figure 8. Achievable receiver signal vs capacitor voltage rating. Higher-rated capacitors permit more turns and thus more coupling; the 2 kV class clears the 3.3 V target.

Recommended point (2 kV class, highest coupling): 119 turns per coil gives $M = 0.0525 \mu\text{H}$, $L = 2.54 \text{ mH}$, and a secondary-tuned RX signal of $\sim 4.1 \text{ V}$ — the strongest coupling among standard capacitor classes and comfortably above the 3.3 V target. If a smaller/cheaper capacitor is required, drop to the 1 kV or 630 V row and accept a proportionally smaller (still very usable) signal.

6 Component Ratings (Recommended Design)

All values are for the recommended 119-turn, 70 mm, AWG16 air coils at 25 kHz with a 5 A peak primary drive.

Component	Value	Rating / notes
Coil self-inductance L	2.54 mH (each)	identical TX and RX coils
Coil AC resistance R	0.534 Ω @25 kHz	skin effect included; use litz for higher Q
Coil Q factor	≈ 100 (assumed)	solid-wire air coil; litz can reach 200+
Mutual inductance M	0.0525 μ H	baseline coupling (no wheel)
Primary series capacitor C_p	16.0 nF	≥ 2 kV peak, ≥ 5 A RMS, polypropylene film, low-ESR
Secondary tuning capacitor C_s	16.0 nF	parallel-tune the RX; ~ 250 V film is ample (low current)
Primary drive current	5.0 A peak	3.5 A RMS through the winding
Resonant drive voltage	2.67 V peak	half-bridge / class-D at 25 kHz
Copper loss per coil	6.7 W	AWG16 handles 5 A; keep coils ventilated
Wire length per coil	26 m	well within a 500 m budget

Why the capacitor is the critical part. At resonance the inductor and capacitor exchange the full reactive energy, so the capacitor sees the coil's peak voltage (up to 2 kV here) while carrying the full 5 A. Use a purpose-built AC power film capacitor (polypropylene, e.g. WIMA FKP / Celeem class); ordinary ceramic or electrolytic parts will fail. The secondary capacitor is easy — it only handles the small sense-coil current.

7 Coil Construction Guide

A practical recipe for winding one coil (make two identical coils, TX and RX):

- Former.** Use a non-magnetic, non-conductive circular former ~ 70 mm outside diameter (3D-printed PLA/ABS, acrylic, or a phenolic ring). A shallow channel ~ 10 – 15 mm wide holds the winding. Never use a steel or aluminium former — it would add its own eddy losses and mask the wheel.
- Wire.** AWG16 (1.3 mm) enamelled copper magnet wire for the ~ 5 A drive. For a higher-Q, cooler coil, use litz wire of equivalent copper area — it cuts AC resistance at 25 kHz. You need ~ 30 m per coil.
- Turns.** Wind **119 turns** in a tidy multi-layer solenoid, all in the same direction, keeping tension even. Count carefully — coupling scales as turns $\times$ turns, so a miscount on either coil shifts the tuning. Anchor the start and finish with tape or a dab of epoxy.
- Termination.** Scrape/tin the enamel off both ends. Bring out a twisted pair to minimise pickup. Keep the two coil leads short and identical in length between TX and RX for symmetry.
- Primary tuning.** Put the ~ 16 nF / 2 kV film capacitor in **series** with the TX coil to form a series-resonant tank at 25 kHz ($f = 1 / (2\pi\sqrt{L \cdot C})$). Trim with a small parallel cap to hit resonance (watch for peak current at fixed drive voltage).
- Secondary tuning.** Put a ~ 16 nF capacitor in **parallel** with the RX coil to resonate it at the same 25 kHz. This multiplies the received voltage by the coil Q (~ 100), turning ~ 40 mV of raw coupling into several volts. Keep the preamplifier input impedance high so it does not load down the Q.
- Driver.** Drive the primary tank from a half-bridge / class-D stage at 25 kHz delivering ~ 5 A peak (only ~ 2.7 V across the tuned tank, so a low-voltage supply suffices). A current-sense/AGC loop keeps the baseline steady so the wheel dip is measured against a stable reference.

8. **Mounting & detection.** Fix TX and RX symmetrically across the rail so a wheel passing over the rail head moves through the coupling region. Rectify/envelope-detect the RX amplitude and threshold it: a drop past ~30–50% of baseline is one axle. Two coil sets spaced along the rail give direction and speed.

Tuning check. With no wheel, adjust each tank for maximum, stable RX amplitude at 25 kHz. Then pass a steel plate/wheel through: you should see the amplitude collapse by the ~90% the FEM predicts and recover cleanly. That swing, not the absolute level, is what the counter thresholds.

8 Reproducibility & Code

Every figure and number here is regenerated by scripts in the project folder: `optimize_air_core.py` (design table + ratings), `femm_wheel_dip.py` (the live wheel-dip FEM run via `run_wheel.bat`), `femm_sweep.py` (property-based DOE), and `make_figs.py` (plots). All FEMM scripts now work on scratch copies, so the base model is never modified. Analytical models run in any Python 3 + NumPy environment; FEMM scripts need a Windows FEMM install. See `IMPROVEMENTS.md` for the full change log.